

MULTIPLE CHOICE QUESTIONS

UNIT-I & II

1. Basics & Definition of Operating System

1. An Operating System is primarily responsible for:

- A. Compiling programs
- B. Managing hardware and software resources
- C. Designing hardware
- D. Editing documents

2. Which of the following best defines an Operating System?

- A. A hardware component
- B. A system software that acts as an interface between user and hardware
- C. An application software
- D. A programming language

3. The main goal of an Operating System is to:

- A. Increase program size
- B. Simplify hardware design
- C. Provide efficient and convenient use of resources
- D. Eliminate hardware

2. Generations of Operating Systems

4. First-generation operating systems were mainly based on:

- A. Multiprogramming
- B. Batch processing
- C. Time sharing
- D. Real-time processing

5. Which generation introduced multiprogramming?

- A. First
- B. Second
- C. Third
- D. Fourth

6. Time-sharing systems became popular in which generation of OS?

- A. First
- B. Second
- C. Third
- D. Fifth

3. Types of Operating Systems

7. Which OS allows multiple users to access the system simultaneously?

- A. Single-user OS
- B. Batch OS
- C. Multi-user OS
- D. Embedded OS

8. Real-time operating systems are mainly used in:

- A. Word processing

- B. Online banking
- C. Embedded and control systems
- D. Gaming

9. Which of the following is NOT a type of operating system?

- A. Distributed OS
- B. Network OS
- C. Compiler OS
- D. Real-time OS

4. OS Services & System Calls

10. Which service of OS helps in preventing unauthorized access?

- A. File management
- B. Security
- C. Process management
- D. I/O management

11. System calls are used to:

- A. Compile a program
- B. Access hardware directly
- C. Request services from the OS
- D. Write device drivers

12. Which of the following is NOT an OS service?

- A. Program execution
- B. Error detection
- C. Memory compilation
- D. File system manipulation

5. Virtual Machine Concept

13. A virtual machine is:

- A. A physical machine
- B. An application program
- C. A software emulation of a physical computer
- D. A compiler

14. The main advantage of virtual machines is:

- A. Reduced security
- B. Hardware dependency
- C. Isolation and resource sharing
- D. Slower execution

6. Processes

15. A process is best defined as:

- A. A program stored on disk
- B. A program in execution
- C. A compiled code
- D. An OS service

16. Which of the following represents a process state?

- A. Compile
- B. Ready

- C. Install
- D. Execute file

17. Parent and child processes are examples of:

- A. Independent processes
- B. Cooperating processes
- C. Deadlocked processes
- D. Orphan processes

7. Process States & PCB

18. Which process state indicates the process is waiting for I/O?

- A. Ready
- B. Running
- C. Waiting
- D. New

19. Process Control Block (PCB) contains:

- A. Program code only
- B. CPU registers and process state
- C. Only memory information
- D. Only scheduling algorithm

20. Which of the following is NOT stored in PCB?

- A. Process ID
- B. Program counter
- C. Scheduling information
- D. Source code

8. CPU & Process Scheduling

21. CPU scheduling is concerned with:

- A. Memory allocation
- B. Process selection for execution
- C. File access
- D. Disk formatting

22. Which scheduling is used when CPU is taken away forcibly from a process?

- A. Non-Preemptive
- B. Cooperative
- C. Preemptive
- D. Batch

23. FCFS scheduling may suffer from:

- A. Starvation
- B. Deadlock
- C. Convoy effect
- D. Pre-emption

9. Scheduling Algorithms

24. Shortest Job First (SJF) scheduling minimizes:

- A. Turnaround time
- B. Waiting time

C. Response time

D. Throughput

25. Round Robin scheduling is mainly designed for:

A. Batch systems

B. Real-time systems

C. Time-sharing systems

D. Embedded systems

26. The key parameter in Round Robin scheduling is:

A. Arrival time

B. Burst time

C. Time quantum

D. Priority

10. Scheduling Objectives & Criteria

27. The main objective of process scheduling is to:

A. Maximize CPU idle time

B. Minimize system cost

C. Optimize CPU utilization

D. Increase hardware size

28. Which scheduler selects processes from ready queue?

A. Long-term scheduler

B. Medium-term scheduler

C. Short-term scheduler

D. Dispatcher

29. Which is NOT a scheduling criterion?

A. CPU utilization

B. Throughput

C. Turnaround time

D. File size

30. Starvation occurs when:

A. CPU is idle

B. Process never gets CPU time

C. Memory is full

D. Deadlock happens